

We now consider the **Yang–Nevzorov model** based on full observations X_t with:

$$X_t \sim F^{\gamma^t}, \quad t = 1, \dots, T$$

and F is Fréchet (A, α) .

✓ 1. Distribution under Yang–Nevzorov

If $X_t \sim F^{\gamma^t}$, then:

- CDF:

$$F_t(x) = [F(x)]^{\gamma^t}$$

- Density:

$$f_t(x) = \gamma^t f(x) F(x)^{\gamma^t - 1}$$

✓ 2. Likelihood (all observations)

$$\mathcal{L}(\gamma, A, \alpha) = \prod_{t=1}^T \gamma^t \cdot f(x_t) \cdot F(x_t)^{\gamma^t - 1}$$

✓ 3. Log-likelihood

$$\ell = \sum_{t=1}^T [t \log \gamma + \log f(x_t) + (\gamma^t - 1) \log F(x_t)]$$

✓ Substitute Fréchet

Recall:

$$\log f(x) = \log(A\alpha) - (1 + \alpha) \log(Ax) - (Ax)^{-\alpha}$$

$$\log F(x) = -(Ax)^{-\alpha}$$

✓ Final log-likelihood

Define:

$$z_t = (Ax_t)^{-\alpha}, \quad \log_t = \log(Ax_t)$$

$$\begin{aligned} \ell(\gamma, A, \alpha) = & \left(\sum_{t=1}^T t \right) \log \gamma \\ & + T \log(A\alpha) \\ & - (1 + \alpha) \sum_{t=1}^T \log(Ax_t) \\ & - \sum_{t=1}^T z_t \\ & - \sum_{t=1}^T (\gamma^t - 1) z_t \end{aligned}$$

✓ Simplified

$$\ell = S_1 \log \gamma + T \log(A\alpha) - (1 + \alpha) \sum \log(Ax_t) - \sum w_t z_t$$

where:

$$w_t = \gamma^t$$

✓ 4. First-order derivatives

◆ w.r.t. A

$$\frac{\partial \ell}{\partial A} = -\frac{\alpha T}{A} + \alpha \sum_t \gamma^t x_t (Ax_t)^{-\alpha-1}$$

◆ w.r.t. α

$$\frac{\partial \ell}{\partial \alpha} = \frac{T}{\alpha} - \sum \log(Ax_t) + \sum \gamma^t z_t \log(Ax_t)$$

◆ w.r.t. γ

$$\frac{\partial \ell}{\partial \gamma} = \sum_t \frac{t}{\gamma} - \sum_t t \gamma^{t-1} z_t$$

✓ 5. Second-order derivatives

◆ AA

$$\ell_{AA} = \frac{\alpha T}{A^2} - \alpha(\alpha + 1) \sum \gamma^t x_t^2 (Ax_t)^{-\alpha-2}$$

◆ $\alpha\alpha$

$$\ell_{\alpha\alpha} = -\frac{T}{\alpha^2} - \sum \gamma^t z_t (\log(Ax_t))^2$$

◆ $A\alpha$

$$\ell_{A\alpha} = -\frac{T}{A} + \sum \gamma^t x_t (Ax_t)^{-\alpha-1} (1 - \alpha \log(Ax_t))$$

◆ $\gamma\gamma$

$$\ell_{\gamma\gamma} = -\sum \frac{t}{\gamma^2} - \sum t(t-1)\gamma^{t-2}z_t$$

◆ Cross derivatives

$A\gamma$

$$\ell_{A\gamma} = \alpha \sum t\gamma^{t-1}x_t(Ax_t)^{-\alpha-1}$$

$\alpha\gamma$

$$\ell_{\alpha\gamma} = \sum t\gamma^{t-1}z_t \log(Ax_t)$$